Operation Management

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<MBAZG526/QMZG526, Operation Management>
Lecture No. 6

Class Discussion

- You have an individual 2 Bedroom House where you live in with your family.
- You now need a 3 Bedroom House.
- Money is not a constraint.
- How you achieve your goal of 3 Bedroom House.

Strategic Location Decision

- The addition of new location in the existing system (expanding the facilities in the old site).
- The addition of new location somewhere else and maintaining the old site.
- Closing of existing facilities and move towards another location.

Strategic Location Decision

Oil Plant and Sea Transportation

Source: Operation Management by William J Stevenson

Factors that affect Location Decision

- Country.
- Region.
- Community.

Factors that affect Identifying Country

- Government.
- Cultural and economic issues.
- Location of markets.
- Labour .
- Tax rates, Interest rates, Inflation rates, Financial incentives, Exchange rates and currency risk.
- Customer Preferences.
- Safety.

Factors that affect Identifying Region

- Corporate desires
- Attractiveness of region (culture, taxes, climate, etc.)
- Labour availability, costs, attitudes toward unions.
- Cost and availability of utilities.
- Government incentives and fiscal policies.
- Proximity to raw materials and customers.
- Land/construction costs.

Factors that affect Identifying Community

- Site size and cost.
- Transportation system.
- Zoning restrictions.
- Proximity of services/supplies needed.
- Customer density and demographics.
- Proximity to Competitors (Clustering).

Service Location Strategy

- Manufacturing -sector focuses on minimizing cost.
- Service sector focuses on maximizing revenue.
- Clustering plays a huge role in Service Location Strategy.

OM in Action How La Quinta Selects Profitable Hotel Sites

One of the most important decisions a lodging chain makes is location. Those that pick good sites more accurately and quickly than competitors have a distinct advantage. La Quinta Inns, headquartered in San Antonio, Texas, is a moderately priced chain of 800 inns. To model motel selection behavior and predict success of a site, La Quinta turned to regression analysis.

The hotel started by testing 35 independent variables, trying to find which of them would have the highest correlation with predicted profitability, the dependent variable. Variables included: the number of hotel rooms in the vicinity and their average room rates; local attractions such as office buildings and hospitals that drew potential customers to a 4-mile-radius trade area; local population and unemployment rate; the number of inns in a region; and physical characteristics of the site, such as ease of access or sign visibility.

In the end, the regression model chosen, with an R^2 of 51%, included four predictive variables: (1) the price of the inn, (2) median income levels, (3) the state population per inn, and (4) the location of nearby colleges (which serves as a proxy for other demand generators). La Quinta then used the regression model to predict profitability and developed a cutoff that gave the best results for predicting success or failure of a site. A spreadsheet is now used to implement the model, which applies the decision rule and suggests "build" or "don't build." The CEO likes the model so much that he no longer feels obliged to personally select new sites.

Mike Booth/Alamy

Sources: S. Kimes and J. Fitzsimmons, *Interfaces* 20, no. 2: 12–20; and G. Keller, *Statistics for Management and Economics*, 8th ed. Cincinnati-Cengage, 2008: 679.

TABLE 8.6 Location Strategies—Service vs. Goods-Producing Organizations

SERVICE/RETAIL/PROFESSIONAL	GOODS-PRODUCING
REVENUE FOCUS	**COST FOCUS**
Volume/revenue Drawing area; purchasing power Competition; advertising/pricing **Physical quality** Parking/access; security/lighting; appearance/image **Cost determinants** Rent Management caliber Operation policies (hours, wage rates)	**Tangible costs** Transportation cost of raw material Shipment cost of finished goods Energy and utility cost; labor; raw material; taxes, and so on **Intangible and future costs** Attitude toward union Quality of life Education expenditures by state Quality of state and local government
TECHNIQUES	**TECHNIQUES**
Regression models to determine importance of various factors Factor-rating method Traffic counts Demographic analysis of drawing area Purchasing power analysis of area Center-of-gravity method Geographic information systems	Transportation method Factor-rating method Locational cost–volume analysis Crossover charts
ASSUMPTIONS	**ASSUMPTIONS**
Location is a major determinant of revenue High customer-interaction issues are critical Costs are relatively constant for a given area; therefore, the revenue function is critical	Location is a major determinant of cost Most major costs can be identified explicitly for each site Low customer contact allows focus on the identifiable costs Intangible costs can be evaluated

Source: Heizer, Render, Muckson

Any Question

Techniques for Location Selection

- Cost Volume Analysis.
- Factor Rating Method.
- Centre of Gravity Method.
- Transportation Model.

Cost Volume Analysis

- Determine the fixed and variable cost for each location.
- Plot the costs for each location, with costs on the vertical axis of the graph and annual volume on the horizontal axis.
- Select the location that has the lowest total cost for the expected production volume.

Class Exercise

- Esmail Mohebbi, owner of European Ignitions Manufacturing, needs to expand his capacity. He is considering three locations—Athens, Brussels, and Lisbon—for a new plant.
- The company wishes to find the most economical location for an expected volume of 2,000 units per year.
- The fixed costs per year at the sites are $30,000, $60,000, and $110,000, respectively; and variable costs are $75 per unit, $45 per unit, and $25 per unit, respectively. The expected selling price of each ignition system produced is $120.

Solution

For Athens:

Total cost = 30,000 + 75(2,000) = 180,000

For Brussels:

Total cost = 60,000 + 45(2,000) = 150,000

For Lisbon:

Total cost = 110,000 + 25(2,000) = 160,000

Profit for Brussels :

Total revenue - Total cost = 120(2,000) - 150,000 = 90,000 per year

Class Exercise

- Esmail Mohebbi, owner of European Ignitions Manufacturing, needs to expand his capacity. He is considering three locations—Athens, Brussels, and Lisbon—for a new plant.
- The fixed costs per year at the sites are $30,000, $60,000, and $110,000, respectively; and variable costs are $75 per unit, $45 per unit, and $25 per unit, respectively.

- The crossover point for Athens and Brussels is:

$30{,}000 + 75(x) = 60{,}000 + 45(x)$

$30(x) = 30{,}000$

$x = 1{,}000$

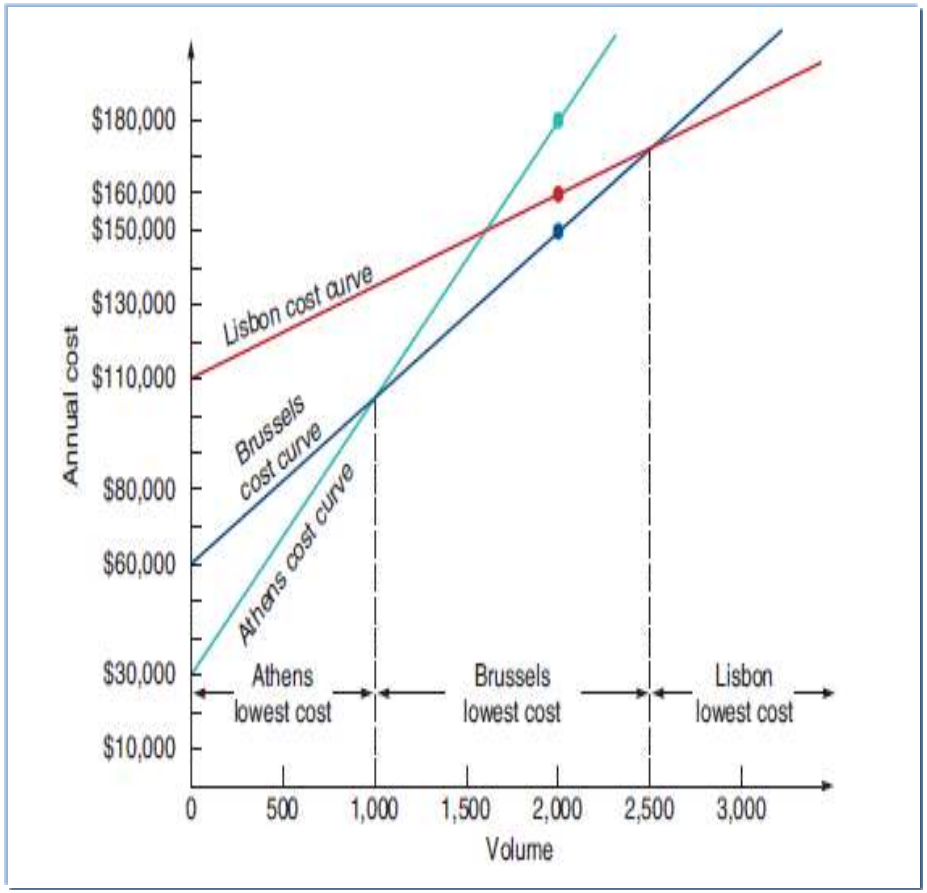
- And the crossover point for Brussels and Lisbon is:

$60{,}000 + 45(x) = 110{,}000 + 25(x)$

$20(x) = 50{,}000$

$x = 2{,}500$

- For x < 1000, Select Athens.
- For x > 2500, Select Lisbon.
- 1000 < x > 2500, Select Brussels.
- Concept of Cross Over Chart.



Source: Heizer, Render, Muckson

Factor Rating Method

- Develop a list of relevant factors called *key success factors*
- Assign a weight to each factor to reflect its relative importance in the company's objectives.
- Develop a scale for each factor (for example, 1 to 10 or 1 to 100 points).
- Multiply the score by the weights for each factor and total the score for each location.
- Make a recommendation based on the maximum point score, considering the results of other quantitative approaches as well.

Class Exercise

- Five Flags over Florida, a U.S. chain of 10 family-oriented theme parks, has decided to expand overseas by opening its first park in Europe. It wishes to select between France and Denmark.

TABLE 8.4 Weights, Scores, and Solution

KEY SUCCESS FACTOR	WEIGHT	SCORES (OUT OF 100)		WEIGHTED SCORES	
		FRANCE	DENMARK	FRANCE	DENMARK
Labor availability and attitude	.25	70	60	(.25)(70) = 17.50	(.25)(60) = 15.00
People-to-car ratio	.05	50	60	(.05)(50) = 2.50	(.05)(60) = 3.00
Per capita income	.10	85	80	(.10)(85) = 8.50	(.10)(80) = 8.00
Tax structure	.39	75	70	(.39)(75) = 29.25	(.39)(70) = 27.30
Education and health	.21	60	70	(.21)(60) = 12.60	(.21)(70) = 14.70
Totals	1.00			70.35	68.00

Source: Heizer, Render, Muckson

Any Question

Centre Of Gravity Method

- **Centre of gravity method :** Method for locating a distribution centre that minimizes distribution cost.

$$x\text{-coordinate of the center of gravity} = \frac{\sum_i x_i Q_i}{\sum_i Q_i}$$

$$y\text{-coordinate of the center of gravity} = \frac{\sum_i y_i Q_i}{\sum_i Q_i}$$

where x_i = x-coordinate of location i
y_i = y-coordinate of location i
Q_i = Quantity of goods moved to or from location i

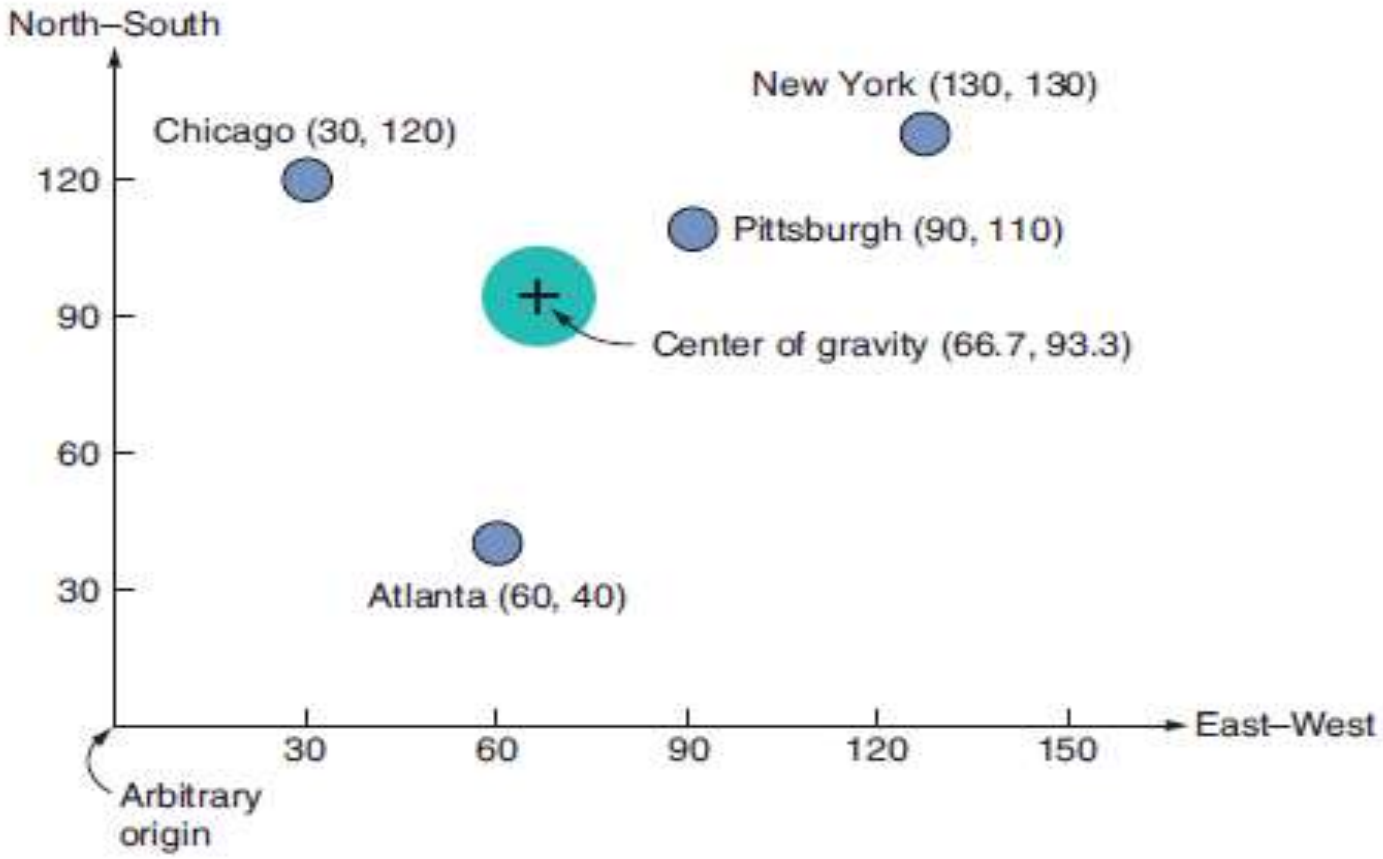
Source: Heizer, Render, Muckson

Class Exercise

- Quain's Discount Department Stores, a chain of four large Target-type outlets, has store locations in Chicago, Pittsburgh, New York, and Atlanta.
- They are currently being supplied out of an old and inadequate warehouse in Pittsburgh, the site of the chain's first store.
- The firm wants to find some "central" location in which to build a new warehouse.

TABLE 8.5 Demand for Quain's Discount Department Stores

STORE LOCATION	NUMBER OF CONTAINERS SHIPPED PER MONTH
Chicago	2,000
Pittsburgh	1,000
New York	1,000
Atlanta	2,000



Source: Heizer, Render, Muckson

$$= \frac{(30)(2000) + (90)(1000) + (130)(1000) + (60)(2000)}{2000 + 1000 + 1000 + 2000} = \frac{400{,}000}{6{,}000}$$

$$= 66.7$$

y-coordinate of the center of gravity:

$$= \frac{(120)(2000) + (110)(1000) + (130)(1000) + (40)(2000)}{2000 + 1000 + 1000 + 2000} = \frac{560{,}000}{6{,}000}$$

$$= 93.3$$

Source: Heizer, Render, Muckson

Geographic Information System (GIS)

- A system that stores and displays information that can be linked to a geographic location.
- A computer based tool for collecting, storing, retrieving, and displaying demographic data on maps.

Use of GIS

- Logistics companies use GIS data to plan fleet activities.
- Publishers of magazines and newspapers use a GIS to analyze circulation and attract advertisers.
- Real estate companies rely heavily on a GIS to make maps available online to prospective home and business buyers.
- Banks use a GIS to help decide where to locate branch banks.
- Insurance companies use a GIS to determine premiums based on population distribution, etc.

Take Home Assignment

- Read Other Examples of Usage of GIS Services in your text book " Operation Management by Jay Heizer, Barry Render & Chuck Munson—Chapter 8. "

Any Question

Transportation Model

- The objective of the transportation model is to determine the best pattern of shipments from several points of supply (sources) to several points of demand (destinations) so as to minimize total production and transportation.
- A technique for solving a class of linear programming problems.
- Can be solved by using Excel Solver.

Class Exercise

Origin	Plant	Three-Month Production Capacity (units)
1	Cleveland	5,000
2	Bedford	6,000
3	York	2,500
	Total	13,500

The firm distributes its generators through four regional distribution centers located in Boston, Chicago, St. Louis, and Lexington; the three-month forecast of demand for the distribution centers is as follows:

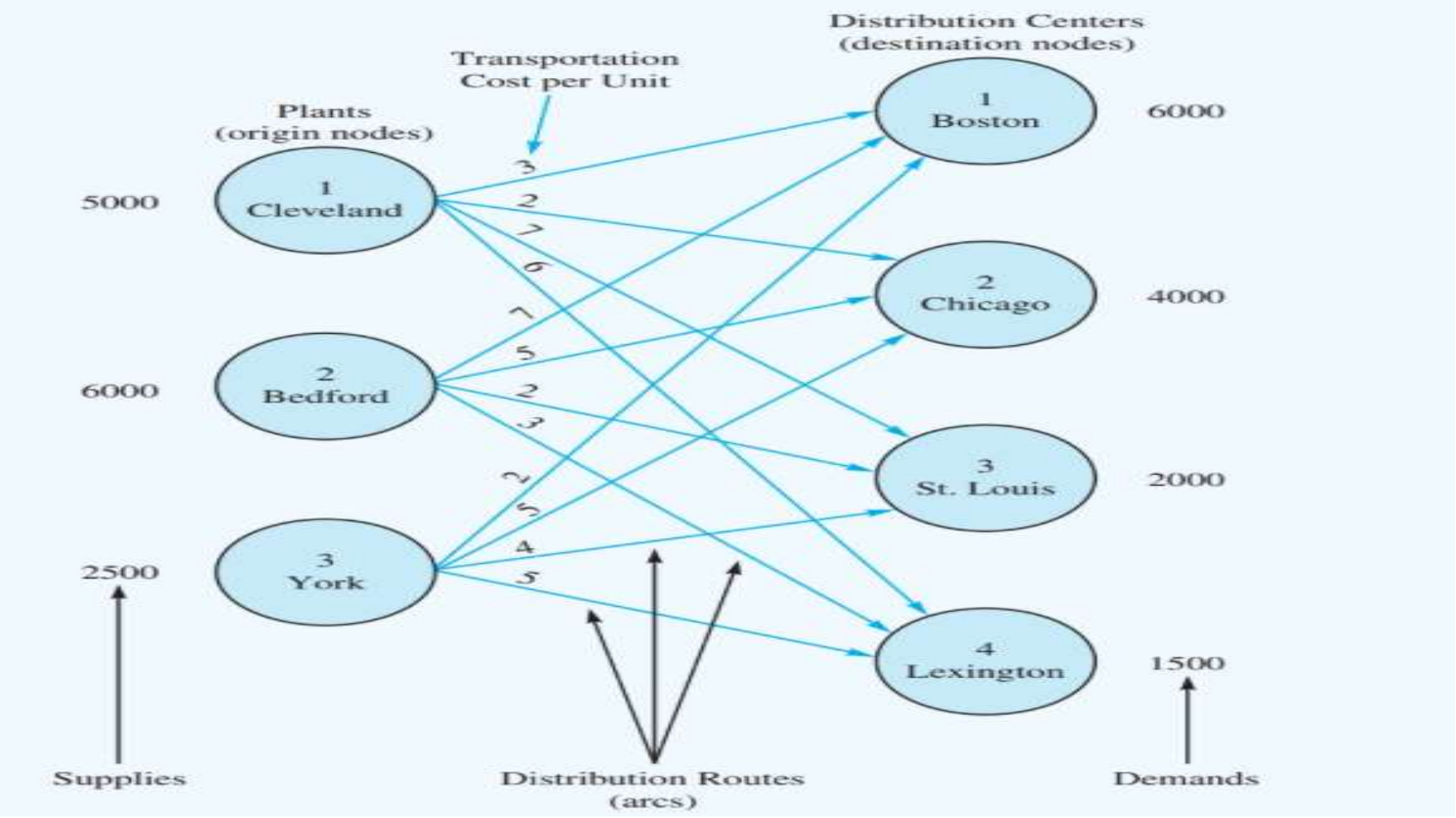
Source: Andersson, Williams

Class Exercise

Destination	Distribution Center	Three-Month Demand Forecast (units)
1	Boston	6,000
2	Chicago	4,000
3	St. Louis	2,000
4	Lexington	1,500
	Total	13,500

Management would like to determine how much of its production should be shipped from each plant to each distribution center. Figure 6.1 shows graphically the 12 distribution routes Foster can use. Such a graph is called a **network**; the circles are referred to as **nodes**

Source: Andersson, Williams



Source: Andersson, Williams

Solution

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management

		Destination									Min Cost	
Origin		Boston	Chicago	St Louis	Lexington			Supply		Z		
	Cleaveland	3	2	7	6			5000				
	Bedford	7	5	2	3			6000				
	York	2	5	4	5			2500				
	Demand	6000	4000	2000	1500							
		Destination										
Origin		Boston	Chicago	St Louis	Lexington	Total		Supply				
	Cleaveland					0	<=	5000				
	Bedford					0	<=	6000				
	York					0	<=	2500				
Total		0	0	0	0							
		=	=	=	=							
	Demand	6000	4000	2000	1500							

Take Home Exercise

- Solved Problem 8.1
- Solved Problem 8.2
- Solved Problem 8.3

Any Question

Further Reading / References

1. **Operation Management by Jay Heizer, Barry Render & Chuck Munson—Chapter 8 & 7**
2. Operation Management by William J Stevenson—Chapter 8
3. An Introduction to Management Science –Chapter 6

THANK YOU